Wireless Communication Standards in Industry: Wi-Fi, Zigbee, and 5G Applications in Smart Factories

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***Abstract*—Modern industrial environments increasingly depend on wireless communication to deliver the flexibility, mobility, and scalability demanded by Industry 4.0. This paper surveys three wireless standards widely deployed in industrial settings: IEEE 802.11 (Wi-Fi), IEEE 802.15.4/Zigbee, and 5G New Radio. It examines their architectures, protocol stacks, application domains, and performance characteristics. Using a representative smart factory scenario, the study derives concrete communication requirements and matches each standard to the use cases it best serves. Key performance metrics including latency, throughput, reliability, and energy consumption are evaluated, and the principal challenges of industrial wireless deployment, namely electromagnetic interference, co-channel contention, multipath fading, and cybersecurity, are discussed. The pros and cons of each standard are identified, along with emerging directions such as Wi-Fi 6/7, private 5G campus networks, Time-Sensitive Networking (TSN) integration, and the outlook towards 6G. The results show that heterogeneous multi-technology deployments best satisfy the full range of industrial requirements, and that Wireless TSN represents the most promising path towards fully dependable wireless factory communication.**

***Index Terms*—wireless industrial communication, Wi-Fi, IEEE 802.11, Zigbee, IEEE 802.15.4, 5G, URLLC, Industry 4.0, latency, smart factory, TSN, private network**

I. Introduction

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Industry 4.0 is transforming manufacturing by linking cyber-physical systems, the Industrial Internet of Things (IIoT), cloud computing, and automation into a unified, more efficient production environment [1]. This vision works only when the various systems on the factory floor can exchange data with each other quickly, reliably, and securely.

Traditionally, plants relied on wired fieldbuses such as PROFIBUS or PROFINET for their dependable timing guarantees. However, these systems come with clear downsides: they are costly to install and difficult to reconfigure. Given how adaptable factories need to be today, fixed cabling often gets in the way. Adding or relocating equipment such as Automated Guided Vehicles (AGVs) and modular production cells is far easier without the burden of cables [2].

This is where wireless communication steps in. It cuts the high cost of installation, makes communication possible in hard-to-reach locations, and lets machinery move around freely, which is next to impossible with fixed connections [1]. Wireless therefore does not only reduce expenses but also adds flexibility, making factory operations considerably more dynamic.

At the same time, the industrial radio environment is one of the most hostile for wireless systems. Factory floors contain large amounts of metal and heavy machinery, which create strong electromagnetic interference (EMI) and complicated multipath fading. As a result, signal quality suffers and packet error rates rise, which makes it difficult to meet the strict latency and reliability requirements of industrial control loops [3].

Three wireless standards have become the most widely used in industrial settings, each targeting a different set of requirements:

- **IEEE 802.11 (Wi-Fi)** is well suited to high-bandwidth mobile applications, such as streaming video from AGVs or machine-vision systems, supporting data rates up to 9.6 Gbps with Wi-Fi 6 [4].
- **IEEE 802.15.4 / Zigbee** is a low-power mesh network that can run for years on batteries thanks to its very low energy consumption, making it ideal for the large numbers of small sensors and actuators used to monitor the environment and track assets around a plant [5].
- **5G New Radio (NR)**, the latest cellular standard, offers an Ultra-Reliable Low-Latency Communication (URLLC) service class that achieves sub-millisecond latency and very high reliability, up to $1-10^{-6}$ for motion-control applications. This level of performance was previously achievable only over wired networks, which is why major industrial players are rapidly adopting private 5G campus networks for their most demanding real-time control tasks [6].

The remainder of this paper is structured as follows. Section II reviews how industrial wireless networks have evolved over time. Section III examines the architectures of the three standards. Section IV provides a thorough evaluation covering a representative factory scenario, key performance figures, and the practical challenges of deployment. Section V discusses emerging directions including Wi-Fi 7, private 5G, Wireless TSN, and the 6G outlook. Finally, Section VI summarises the findings and suggests directions for future research.

II. Literature Review

Section author: Jaleel Ur Rehman

A. Early Wireless Sensor Networks

By the early 2000s, researchers had begun to build the foundations of industrial wireless communication. One of the first seminal works was the survey by Akyildiz et al. on Wireless Sensor Networks (WSNs), which established energy efficiency and reliable packet delivery as central design considerations, criteria that are still followed by standards-development bodies today [7]. In 2003, the IEEE standardised 802.15.4, defining the rules for the physical and Medium Access Control (MAC) layers for low-rate, low-power wireless personal area networks (LR-WPANs). The Zigbee Alliance then extended this with network and application layer specifications to provide a complete stack for sensor and actuator applications [5].

In parallel, experimental trials pinpointed the main problem areas: variable medium access, unreliable delay under load fluctuations, and vulnerability to electromagnetic interference from industrial equipment. Among these works, the investigations by Willig showed that factory environments with metallic floors induce strong multipath fading, which causes signal attenuation and distorted transmission [2]. These early findings became the design requirements for modern industrial wireless standards.

B. Industrial Wi-Fi Evolution

Earlier generations of 802.11b/g networking suffered from high and fluctuating latency that made closed-loop operation infeasible. The 802.11e QoS amendment improved this by allowing features such as Enhanced Distributed Channel Access (EDCA) to prioritise traffic. MIMO antennas introduced in 802.11n offered improved bandwidth and capabilities such as spatial multiplexing, which renewed interest in Wi-Fi on the factory floor [8]. A 2025 testbed study showed that Wi-Fi 6, under controlled industrial conditions, can provide latencies below 5 ms [4].

C. 5G and the URLLC Paradigm

The 3GPP Release 15 (2018) standardised the 5G New Radio air interface, while Release 16 (2020) added mini-slot scheduling, enhanced HARQ, and traffic preemption for URLLC in factory automation. For motion control, the targets are a latency of 1 ms or less and a reliability of $1 - 10^{-6}$, while closed-loop process control requires 4 ms or less with a reliability of $1-10^{-8}$ [6]. Industry analyses report that private 5G investment is accelerating rapidly, projected to surpass $5 billion annually by 2028, driven largely by Industry 4.0 deployments [9].

D. Towards Wireless Determinism: TSN Integration

A more recent direction combines wireless technologies with IEEE 802.1 Time-Sensitive Networking (TSN) standards. TSN ensures predictable, low-latency delivery of traffic across Ethernet through scheduling mechanisms such as Credit-Based Shaping (802.1Qav), Time-Aware Shaping (802.1Qbv), and Cyclic Queuing and Forwarding (802.1Qch). The 3GPP specified a bridge function in 5G Release 16 that maps TSN flows over the 5G air interface [10].

As a result, an end-to-end guarantee is achieved from wired Ethernet, through the wireless link, to the device on the floor. This is significant because it allows a robot arm connected over 5G to be driven by the same TSN scheduler that drives a wired EtherCAT axis, making the boundary between wired and wireless industrial systems effectively disappear. The IEEE is also actively developing Wi-Fi TSN standards to support comparable deterministic guarantees over IEEE 802.11 links [11].

III. Wireless Standards: Architecture and Protocol Stacks

Section author: Iko-ojo Victor Odaudu

A. IEEE 802.11 (Wi-Fi)

IEEE 802.11 is one of the most widely known wireless standards, and it operates across the 2.4 GHz, 5 GHz, and (with Wi-Fi 6E) the 6 GHz unlicensed bands [12]. The standard covers two main layers, the Physical layer (PHY) and the Medium Access Control layer (MAC), and it uses Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA) as the way devices take turns accessing the channel. With Wi-Fi 6 (802.11ax), two important improvements were added: Orthogonal Frequency Division Multiple Access (OFDMA), which lets multiple devices share the channel at the same time, and Target Wake Time (TWT), which lets devices schedule when they wake up so there is less congestion on the network [4]. The main 802.11 generations relevant to industrial use are summarised in Table I.

TABLE I
Evolution of Key IEEE 802.11 Standards for Industrial Use

Std.	Name	Max Rate	Key Feature
802.11n	Wi-Fi 4	600 Mbps	MIMO, QoS (EDCA)
802.11ac	Wi-Fi 5	3.5 Gbps	MU-MIMO, beamforming
802.11ax	Wi-Fi 6	9.6 Gbps	OFDMA, TWT, 6 GHz (6E)
802.11be	Wi-Fi 7	46 Gbps	MLO, 4096-QAM

In a factory environment, the access points are usually mounted around the facility and tied into a wired backbone. Equipment such as robots, AGVs, and Human Machine Interfaces (HMIs) all connect through these access points wirelessly. The standard also supports fast roaming through 802.11r, which matters a great deal when a machine is moving across the floor and needs to stay connected without dropouts [8]. Figure 1 shows the IEEE 802.11 protocol stack alongside a typical industrial deployment topology.

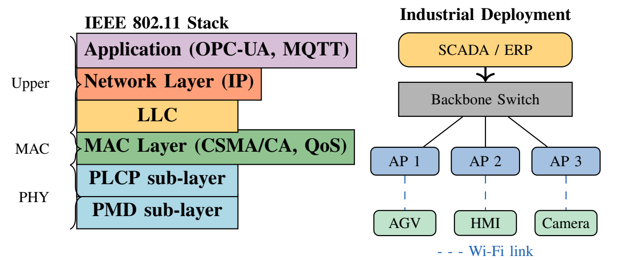

Fig. 1. IEEE 802.11 protocol stack (left) and a representative industrial Wi-Fi deployment with three access points, a backbone switch, a SCADA system, and wireless field devices (right). Figure drawn by the authors.

The main advantages of Wi-Fi are its high data rates, up to 9.6 Gbps with Wi-Fi 6, the very large ecosystem of devices that support it, and the fact that it works natively with standard TCP/IP, so it fits into existing factory IT setups fairly easily. On the other hand, it can become unreliable when the network is under heavy load, and because it uses unlicensed spectrum, it can suffer interference from other sources such as Bluetooth devices or nearby microwave equipment [1].

B. IEEE 802.15.4 / Zigbee

IEEE 802.15.4 is the underlying standard that defines the Physical (PHY) and Medium Access Control (MAC) layers for low-power Wireless Personal Area Networks (WPANs). Zigbee is the most widely used protocol built on top of this, adding the network and application layers needed for mesh communication to work [5]. It was developed specifically for use cases where battery life and cost matter more than speed, which makes it quite different from Wi-Fi in terms of what it is actually trying to achieve.

The standard operates mainly at 2.4 GHz and reaches data rates of up to 250 kbps. It defines three types of devices: the Coordinator, which starts and manages the whole network; Routers, which forward messages between nodes; and End Devices, which are simple low-power nodes that only communicate with their parent device. This arrangement allows Zigbee to support star, tree, and mesh topologies depending on what the application needs [13]. Figure 2 shows the full Zigbee protocol stack and the three supported network topologies.

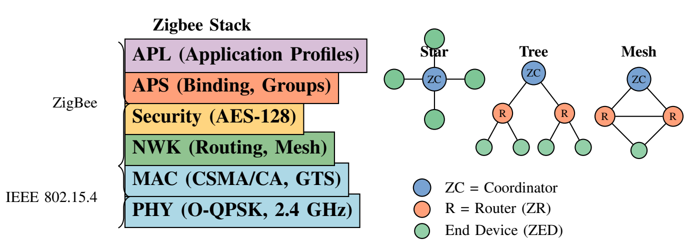

Fig. 2. Zigbee/IEEE 802.15.4 protocol stack (left) and network topologies (right). Figure drawn by the authors based on [5], [13].

In industrial settings, Zigbee is mainly used for sensor networks and monitoring applications where data rates are low but battery life is very important. End Devices can sleep most of the time and only wake up when they need to send data, which is how battery lifetimes of two to five years are achievable even on small cells. This makes it very practical for deploying large numbers of sensors across a factory without running power cables to every single one of them [1].

The main advantages are its very low power consumption, low hardware cost, and a self-healing mesh topology that keeps the network running even if one node fails. It is well suited to dense sensor networks where hundreds of devices need to be deployed cheaply and run for years without much maintenance. The disadvantages are that the data rate is too low for anything real-time, and with a range of only 10 to 100 metres per node, it needs a fair number of routers to cover a large factory floor properly [5].

C. 5G New Radio (NR)

5G New Radio (NR) was standardised by the 3rd Generation Partnership Project (3GPP) starting from Release 15, and it is the newest of the three standards covered in this paper. Unlike Wi-Fi and Zigbee, which both use unlicensed spectrum, 5G typically operates on licensed frequency bands, which gives it more predictable and interference-free performance.

The standard defines three service classes: enhanced Mobile Broadband (eMBB) for high-bandwidth applications, massive Machine Type Communications (mMTC) for large numbers of low-power devices, and Ultra-Reliable Low-Latency Communication (URLLC), which is the most relevant one for industrial use. For motion control, URLLC targets latency below 1 ms and a reliability of $1 - 10^{-6}$, meaning only about one packet in a million is allowed to fail [6]. It also supports flexible subcarrier spacings from 15 kHz up to 240 kHz, which allows slot durations as short as 0.125 ms for time-critical traffic. Figure 3 shows the architecture of a private 5G campus network for an industrial facility.

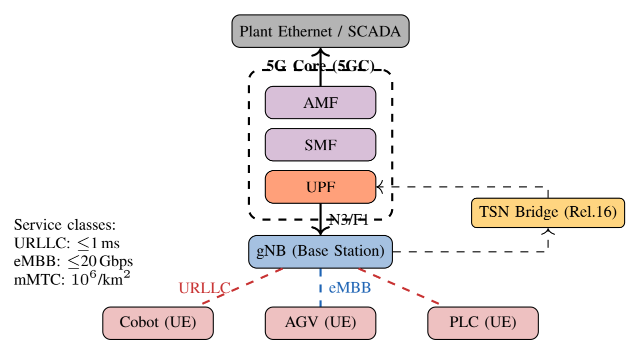

Fig. 3. Private 5G campus network architecture. The on-premises 5G Core and gNB keep data within the facility. A TSN bridge (3GPP Rel. 16) enables end-to-end deterministic scheduling to cobots, AGVs, and PLCs. Figure drawn by the authors based on [6], [11].

For industrial deployments, 5G is usually set up as a private campus network, where the company installs its own base station, called a gNodeB (gNB), and a standalone 5G core locally on site. This keeps all data within the facility and reduces latency, which is especially useful for factories that handle sensitive production data and cannot afford to route it through a public network [9].

From Release 16 onwards, 5G can also act as a Time-Sensitive Networking (TSN) bridge through the 3GPP-defined TSN translator, which enables deterministic scheduling over the wireless link with time-synchronisation accuracy better than one microsecond [11]. This was previously only achievable with wired networks, so it is a significant step for industrial wireless and opens the door for 5G to replace cables in even the most timing-sensitive factory applications.

Special features: URLLC latency and reliability, licensed spectrum, private-network option, and TSN capability. **Limitations:** high cost, high device power consumption, and complex deployment.

IV. Evaluation

Section author: Muhammad Umer Hayat

A. Illustrative Example: Automated Production Cell

Consider a representative production cell built to Industry 4.0 standards. It includes two collaborative robots that need ultra-low latency of 1 ms or less for each movement, with a reliability allowing only about one failure in a million transmissions ($1 - 10^{-6}$). The same cell also contains twenty-four sensors measuring quantities such as temperature, vibration, and gas concentration; these devices send small data packets every thirty seconds while running on batteries for two years or more. In addition, three AGVs each need a 10 Mbps connection with latency not exceeding 10 ms, and a machine-vision station needs to transfer 5 MB image files to the edge server within 0.5 s. No single standard can serve all of these needs, which motivates the heterogeneous deployment shown in Figure 4.

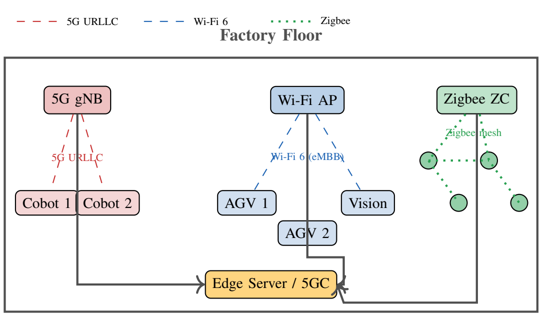

Fig. 4. Heterogeneous wireless deployment for the example smart factory. 5G URLLC connects the cobots; Wi-Fi 6 serves the AGVs and machine-vision station; a Zigbee mesh connects the sensor network. All traffic converges at the edge server. Figure drawn by the authors.

B. Communication Requirements Mapping

The cobots are connected by 5G URLLC, since they have the strictest latency and reliability demands. The sensor network uses Zigbee because of its energy efficiency and long battery life. The AGVs and the vision system use Wi-Fi 6. Table II maps the production-cell requirements to each standard [1].

TABLE II
Communication Requirements vs. Wireless Standard Suitability

Requirement	**Wi-Fi 6**	**Zigbee**	**5G URLLC**
Latency $\leq$ 1 ms	×	×	✓
Latency $\leq$ 10 ms	✓	×	✓
Throughput $\geq$ 10 Mbps	✓	×	✓
Battery life $\geq$ 2 yr	×	✓	×
Mesh (self-healing)	Limited	✓	Limited
Deterministic (TSN)	Emerging	×	Emerging
Licensed spectrum	×	×	✓
Deployment cost	Low	Very Low	High

C. Performance Comparison

The three standards differ sharply across the key metrics. In terms of latency, 5G URLLC reaches as low as 1 ms, Wi-Fi 6 varies between roughly 2 and 5 ms depending on network congestion, and Zigbee adds about 15 to 30 ms per hop. For throughput, Wi-Fi 6 delivers several gigabits per second and 5G NR peaks at around 20 Gbps in eMBB mode, while Zigbee is capped at 250 kbps. On energy use, Zigbee is by far the most efficient, running for years on battery power, whereas 5G devices generally need a continuous power supply. Reliability also differs greatly: 5G URLLC offers guaranteed service levels, while Zigbee degrades sharply under heavy 2.4 GHz congestion [4]–[6]. Figure 5 summarises these differences on a normalised scale.

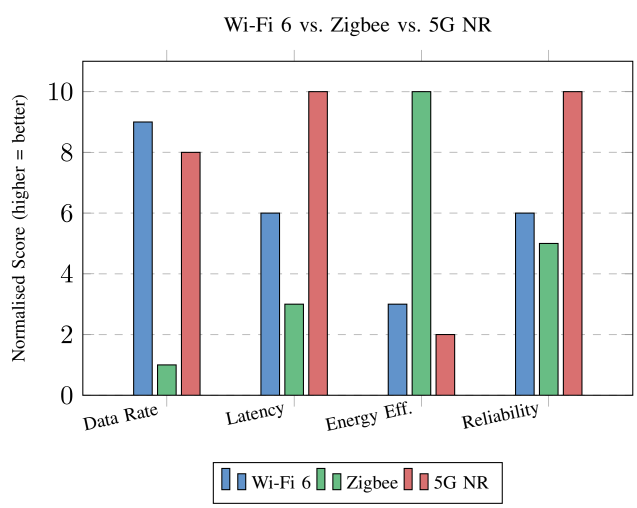

Fig. 5. Normalised performance comparison across four key industrial metrics. Scores derived from published benchmarks [1], [4]–[6]. Figure drawn by the authors.

D. Challenges in Industrial Deployment

Several problems are specific to industrial wireless communication. EMI produced by frequency converters, arc welders, and servo drives can degrade ISM-band wireless systems through broadband interference [3]. In addition, because of the large amount of metal in these environments, multipath fading is common, causing rapid changes in signal characteristics and creating dead zones. OFDM-based technologies such as Wi-Fi 6 and 5G are more resistant to these effects than Zigbee, which uses single-carrier O-QPSK [2].

Co-channel interference must also be considered, since Wi-Fi, Zigbee, and Bluetooth all operate in the 2.4 GHz band. These coexistence issues can be reduced through careful channel allocation and by using higher-frequency bands for Wi-Fi [1]. Cybersecurity is another concern. 5G provides strong built-in security through mutual authentication and end-to-end encryption mandated by 3GPP. Wi-Fi 6 offers good protection through WPA3, provided it is configured correctly. Zigbee uses AES-128 encryption, which is generally sufficient for sensor data, although its key-management system has known limitations and vulnerabilities [5]. Finally, the technologies differ in range: indoor Wi-Fi cells cover about 30 to 50 metres, Zigbee covers roughly 10 to 20 metres per node but extends much further through its mesh, and 5G mmWave covers only tens of metres indoors, requiring dense base-station placement [3].

E. Summary of Pros and Cons

Wi-Fi 6: offers high speed and a large existing device ecosystem. However, it is non-deterministic, operates on shared spectrum, and consumes more energy than Zigbee.

Zigbee: provides low power consumption, inexpensive installation, and self-healing recovery after a node failure. However, it offers a maximum of only 250 kbps throughput, latency up to 30 ms, and may lack sufficient security measures without careful key management.

5G NR: provides all the characteristics needed for URLLC applications, namely ultra-low latency and high reliability, together with licensed spectrum and TSN integration. Its main drawbacks are high deployment cost and complexity.

V. Emerging Trends

Section author: Jaleel Ur Rehman

A. Wi-Fi 7 and Multi-Link Operation

The Multi-Link Operation (MLO) feature of IEEE 802.11be (Wi-Fi 7) achieves both packet-level and path diversity at the link layer. As a result, even if a packet is lost on one band, it can be recovered on another band, avoiding the retransmission delay [14]. The IEEE 802.11bn (Wi-Fi 8) amendment reached Draft 1.0 in 2025, and the emerging consensus is that it will prioritise deterministic performance and reliability rather than further raw throughput increases [12].

B. Private 5G Campus Networks

Adoption has accelerated sharply, with industry analyses projecting that annual private 5G investment will surpass $5 billion by 2028, including production-grade deployments at major manufacturers such as Ford and Hyundai [9]. Because private 5G is optimised for deterministic wireless performance, high device density, and security, it fits well as both a competitor to and a complement of industrial Ethernet and Wi-Fi in new factory designs.

C. Wireless Time-Sensitive Networking

With 5G-TSN (3GPP Release 16), a worst-case latency of 1 ms together with sub-microsecond time synchronisation becomes achievable [11]. If Wireless TSN delivers on its promise, it will close the final gap between wireless and wired industrial communication, eventually allowing the same 1 ms latency and sub-microsecond synchronisation guarantees over Wi-Fi or 5G as over wired PROFINET or EtherCAT [10].

D. 6G Outlook

Looking further ahead, 6G (IMT-2030) targets sub-millisecond latency, terahertz (THz) spectrum, and AI-native network management [15]. For industry, it is expected to support fully wireless closed-loop control at reliability levels not currently achievable over any radio link, potentially replacing the last remaining wired industrial connections.

VI. Conclusions

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This paper has surveyed three of the main wireless standards used in industrial automation and Industry 4.0: Wi-Fi (IEEE 802.11), Zigbee (IEEE 802.15.4), and 5G NR. It examined their protocol architectures, their performance, and the problems that arise when deploying them, illustrated with an example of a smart factory in which four different device classes have different communication needs.

The paper shows that no single standard can handle everything that industry requires. 5G URLLC is essential for real-time control, since it delivers very low latency (1 ms or less) and very high reliability ($1 - 10^{-6}$) over a dedicated channel. Wi-Fi 6 suits high-bandwidth mobile tasks, offering gigabit speeds at relatively low setup cost with a large ecosystem of ready-to-use devices. Zigbee remains the best choice for sensors that need years of battery life with almost no maintenance. Real-world deployments will therefore most likely combine all three networks, matching the most suitable technology to each task.

The major challenges are the electromagnetic interference produced by machinery, connectivity problems caused by metallic environments, heavy use of the 2.4 GHz band, and the threat of cyber-attacks on these systems. Encouragingly, innovations such as Wi-Fi 7's Multi-Link Operation, private 5G solutions, and improvements in security protocols address several of these problems. A key benefit of 5G is its built-in security, with end-to-end encryption and mutual authentication integrated into the whole network architecture, whereas Wi-Fi and Zigbee networks rely more heavily on the operator implementing security correctly.

Perhaps the single most important ongoing effort is the development of the Wireless Time-Sensitive Networking specification. It aims to deliver low-latency data transport in both 5G and Wi-Fi environments so that industrial use cases that once relied solely on wired technology can be supported wirelessly. This includes the prospect of a factory free of cabling that can still support motion-control loops operating as reliably as wired systems.

Future research directions include developing a standard approach to industrial wireless benchmarking under realistic EMI conditions, using AI to support smarter channel selection and interference control, ensuring security for URLLC control loops operating in shared environments, and planning the migration of new and existing facilities from 5G to 6G. As 6G research turns into standardisation activity through the 2030s, the combination of terahertz bandwidth, AI-driven management, and intrinsically secure low-latency communication is expected to drive universal, low-latency automation in an entirely wire-free industrial environment.

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